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Metrics and methods for resilience-aware design of process systems: advances and challenges

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To address potential disruptions that arise from internal and external sources, chemical processes need to be designed with the capacity to withstand and recover from disturbances — often referred to as resilience. To integrate resilience capabilities into process design and operations, it is imperative to be cognizant of contributing factors and develop, determine, and incorporate quantitative resilience performance indicators. This paper provides a brief review of recent progress in conceptual frameworks and quantitative metrics for analyzing and designing resilience-aware process systems. Furthermore, key research opportunities in the field of process system resilience, namely the challenges in integrating resilience throughout the life cycle and across the spatiotemporal scales of a process system, are discussed.

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Introduction

The chemical process industry is a critical component of modern society, producing energy, materials, and specialty chemicals that are crucial to daily life. Chemical processes, however, are also extremely vulnerable to unexpected events such as natural disasters [1], social upheavals [2], and pandemics [3]. Furthermore, disruptions can also be triggered by events from within the process plant itself, such as equipment failures, industrial accidents, and pipeline issues [4,5]. The integration of physical and digital systems that enables the remote operation and control of process plants has also added an extra layer of vulnerability, as cyber-attacks represent another source of potential disruption [6,7]. The consequences of a chemical process failure are significant both in terms of human lives and financial losses, as disasters such as the 1984 Bhopal gas tragedy [8] and the 2010 Deepwater Horizon oil spill [9] have extensively reported and studied. Therefore, it is crucial that chemical process plants are designed to handle these disruptions while maintaining low-risk performance.

Resilience can be defined as the ability of a system to minimize damages and return to a steady state after a disturbance [10]. It is an all-encompassing term that includes elements from important concepts such as flexibility [11], safety [12], reliability and maintainability [13], and robustness [14], with an emphasis on seeking available opportunities to adapt and reconfigure into an improved steady state [15]. Readers are directed to the previously mentioned works for more in-depth discussions on each concept included within resilience. Resilience enhancement strategies can be implemented at every stage in the life cycle of a process plant, but may be most effective when integrated as early as possible, that is, during the process design stage [16]. During process design, trade-offs between resilience and other critical objectives (economic, sustainability, etc.) can also be explored to cost-competitive resilience-aware process systems.

In this article, we provide a brief review of recent work addressing the conceptual and quantitative developments and advancements in the area of process system resilience. In particular, frameworks and metrics that have been developed for the resilience-aware design and analysis of process systems will be discussed. Future research challenges will also be briefly presented.

Table 1

Recent works in the literature that discuss process system resilience, robustness, flexibility, or safety. Papers that provide a quantitative metric are also indicated.

Publication	Resilience	Robustness	Flexibility	Safety	Metric
Khan et al. [17]				✓	
Rogers and Ierapetritou [18]		✓	✓		
Ahooyi et al. [19]		✓		✓	
Jain et al. [20]	✓			✓	
Gong and You [21]	✓	✓			✓
Jain et al. [22]	✓				✓
Abimbola and Khan [23]	✓				✓
Moreno-Sader et al. [24]	✓			✓	✓
Nwankwo et al. [25]		✓		✓	
Tian et al. [26]			✓	✓	
Khan et al. [27]				✓	
Namvar and Bamdad [28]	✓		✓		
Chrisandina et al. [29]	✓				
Al-Douri et al. [30]				✓	
Pawar et al. [31]	✓			✓	✓
Al-Douri et al. [32]				✓	✓
Cheng et al. [33]	✓	✓		✓	✓
Zarei et al. [34]	✓			✓	✓

Overview of current research progress

Conceptual frameworks for analyzing resilient systems

The understanding of resilience as the ability of a system to manage disturbances of various types and with different levels of uncertainties is generally accepted within the field of process systems engineering. However, the specific capabilities, enablers, and components contributing to overall system resilience are being developed in silos. Furthermore, different authors use various terms to describe this ability, such as resilience, robustness, flexibility, safety, and so on. Table 1 lists some key publications in the last decade on process system resilience, including papers that provide a quantitative metric for resilience.

There are three general phases to process system resilience that describe the time period around a disruptive event: pre disruption, during disruption, and post disruption [35]. These phases point to three types of resilience strategies: proactive strategies are implemented at the pre-disruption phase, concurrent strategies at the during-disruption phase, and reactive strategies at the post-disruption phase [36]. Within each phase, different resilience capabilities become more relevant, which in turn informs the strategies most appropriate for that phase. Figure 1 showcases which capabilities are most relevant at each phase of the disaster life cycle.

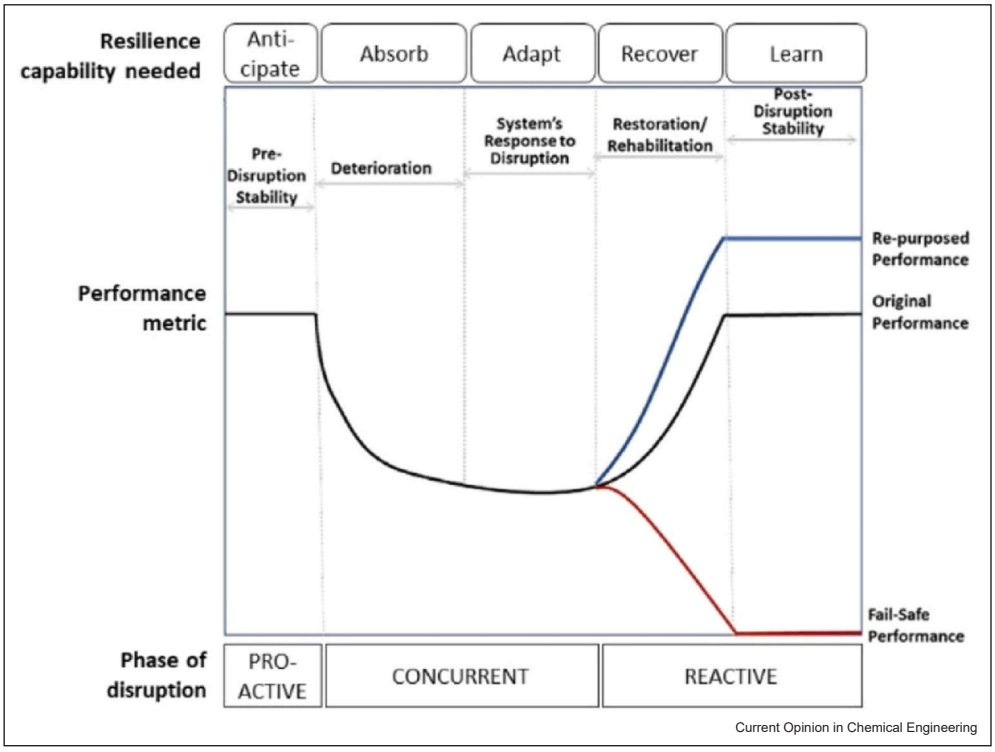
Jain et al. [22] presented the Process Resilience Analysis Framework to integrate the technical and social factors contributing to resilience. This framework relies on early detection, error-tolerant design, recoverability, and plasticity to be applied throughout the life cycle of facilities — design, normal operations, simultaneous operations, and transient operations. Furthermore, the authors developed a Bayesian network-based framework to evaluate the resilience of a process system based on three key contributors: predictability (α), survivability (β), and recoverability (γ). These contributors are in turn dependent on system performance ϕ, which is affected by both internal (θ_i) and external (θ_e) uncertainties. These uncertainties are observed through the use of resilience indicators (R_i). Given the Bayesian network structure, the probability of process system resilience is therefore expressed as a function of probabilities for predictability, survivability, and recoverability.

$$P(\psi|\alpha,\beta,\gamma) = P(R_i) \times P(\theta_e|R_i) \times P(\theta_i|R_i) \times P(\phi|\theta_i,\theta_e) \times P(\alpha|\phi) \times P(\beta|\phi) \times P(\gamma|\phi) \quad (1)$$

A major difficulty in simulating abnormal scenarios is the potential lack of historical data to use as a basis for scenario modeling because some types of disturbances occur very rarely but have significant consequences. For such events, first-principles modeling can be useful to provide meaningful insights and predictions. For example, Kumari et al. [37] developed a parametric reduced-order computational fluid dynamics model based on the *k-nearest-neighbor* machine learning algorithm to estimate the concentration profile of materials released during a rare event. Subsequently, the system can be designed while being cognizant of the impact of rare events.

While most resilience frameworks are designed to respond to one specific disruption, Argyroudis et al. [38] proposed a framework for resilience assessment under multiple hazards. They outlined three ways that multiple hazards can occur: independently, in a cascading

Figure 1



Relevant capabilities of process system resilience at different phases in the disaster life cycle. Adapted from El-Halwagi et al. [15].

manner, and cumulatively. Hazard curves are developed to describe the impact of each hazard at different points of time, and how multiple cascading or cumulative hazards affect the intensity of the event. These curves are then fed into fragility functions that describe how the system is affected by various hazards. Reinstatement and restoration models estimate the recovery time and functionality recovered by the system post hazard. These models are then used to calculate a resilience index. The major benefit of assessing resilience for multiple hazards at once is a more realistic assessment of system resilience and vulnerabilities since most systems are exposed to multiple disruptions at any one time.

Tong et al. [39] published one of the earliest works where a dynamic Bayesian network (DBN) is used for resilience assessment to represent the temporal changes involved in system behavior, namely its ability to withstand disruptions and recover. The authors defined three main contributors to resilience: absorption, adaptation, and restoration, with learning as one of the factors that affects all three. For a specific process system, other external and system factors affecting one or more of these contributors are defined qualitatively. In this work, fuzzy analytic hierarchy process (FAHP) based on expert opinion is used to calculate the conditional probability of each factor, although historical failure and repair rate data can also be used when available. These conditional probabilities are then used to instantiate the DBN and yield a resilience value over time.

Zinetullina et al. [40] built upon the use of a DBN for resilience assessment by using the functional resonance analysis method (FRAM) to qualitatively model how a complex system functions considering socio-technological interactions. The authors outlined a model that utilizes FRAM analysis to model how different operational functions in a process system are affected by each other. Aspen Hysys is used to simulate the process and derive the probability of failure for different units in the system. Both the FRAM analysis and probability of failure results are then fed into a DBN to generate the conditional probabilities for each node on the DBN.

Quantitative metrics for resilience

To accurately incorporate resilience into process design, a quantitative metric that measures system resilience needs to be developed. However, the fact that resilience can be conceptualized through multiple perspectives has also meant that there are many

metrics that have been proposed to quantify various aspects of resilience. Broadly, relevant metrics can be categorized into two types: performance metrics or structural metrics [29,41]. When resilience is viewed as the specific response of a system toward a set of disturbance scenarios, performance metrics such as production level or time taken for recovery are used as proxies to gauge how quickly the system responds and recovers from the given scenarios. Structural metrics, on the other hand, rely on inherent system properties that can be quantified, regardless of the presence of a disturbance scenario.

Performance metrics rely on modeling system behavior during a set of disruption scenarios, so their usefulness greatly depends on the accuracy of the system model used as well as the disruption scenarios chosen for analysis. A risk in using only performance metrics to gauge the resilience of a design is that the system will behave poorly under disruption scenarios that are vastly different from the scenario set chosen to generate the design. However, a major benefit of these metrics is that a detailed model of the system is not necessary as long as its behaviors can be modeled. This allows for simpler models to be used to represent the system.

Gong and You [21] proposed a generalized framework to incorporate resilience metrics and enhancement strategies for resilient-aware process design and operations. The developed optimization model includes three stages: (1) to determine optimal network configuration, equipment capacities, and capital costs; (2) to evaluate the worst-case realization of the availability of process units; and (3) to determine the optimal number of available processes and their corresponding operating levels. The resilience metric employed for the analysis is the ratio between process performance under disruption ($F(t)$) and performance under nominal conditions ($F(t_0)$) over a specified time period ($[t_0, t_n]$):

$$Resilience = \frac{\int_{t_0}^{t} F(t)dt}{\int_{t_0}^{t_n} F(t_0)dt} \tag{2}$$

Castillo-Borja et al. [42] proposed a safety resilience index (SRI) to evaluate the probability of risk-averse operation of processes under unexpected situations. The SRI represents the proportion of incidents that were successfully prevented via safety devices or actions, and is bounded between [0,1]. The developed methodology entails (i) identifying potential hazardous scenarios, (ii) identifying cause–consequence scenarios, and (iii) applying MC simulations for each scenario to generate the probability of successful prevention of hazardous scenarios.

Taleb-Berrouane and Khan [43] introduced dynamic resilience modeling to quantify resilience under varying conditions that impact the system and its performance. The approach was demonstrated through a case study concerning pipeline corrosion. Stochastic Petri Networks (SPNs) were employed to model the system and encapsulate the dynamic processes influencing the occurrence, control, and mitigation of corrosion. The dynamic resilience is calculated as an aggregate of the system's absorptive, adaptive, and restorative capacities during the time of disruption and recovery (T_n)

$$\text{Absorptive capacity(DAB)} = \frac{\int_{t_e}^{t_c} S(t)dt - \int_{t_e}^{t_c} M(t)dt}{\int_{t_e}^{t_c} W(t)dt} \tag{3}$$

$$\text{Adaptive capacity (DAD)} = \frac{\int_{t_c}^{t_r} \mathrm{S(t)dt} - \int_{t_c}^{t_r} \mathrm{M(t)dt}}{\int_{t_c}^{t_r} \mathrm{W(t)dt}} \tag{4}$$

$$\text{Restorative capacity (DRD)} = \frac{\int_{t_r}^{t_t} \mathrm{S(t)dt}}{\int_{t_r}^{t_c} \mathrm{W(t)dt}} \tag{5}$$

$$\text{Dynamic resilience} = \frac{T_n + DAB\Delta T_{ce} + DAD\Delta T_{rc} + DRS\Delta T_{tr}}{T_n + \Delta T_{ce} + \Delta T_{rc} + \Delta T_{tr}} \tag{6}$$

where W(t) is the system performance curve, and $S(t)$ and $M(t)$ are system performances during servicing and monitoring scenarios. t_e represents the time that disruption mitigation measures take effect, t_c is the time for equipment replacement, and t_r is the time at which the system is restored.

Sun et al. [44] used a system-theoretic accident model and processes (STAMP) model to represent a process system. STAMP represents systems as a combination of dependent subsystems and components, and uses control feedback loops to maintain dynamic equilibrium between subsystems. The use of STAMP allows for representation of how subsystems interact with one another and drive system response to a disruption. Therefore, the resilience performance of the whole system $F(t)$ is composed of the sum of its subsystems. This resilience performance is then used to calculate the resilience ratio as per Eq. (2)

$$F(t) = \left[\sum_i \frac{\sum_{j=1}^{mi} s_{nmi}(t)}{mi}\right]/i \tag{7}$$

where i is the number of subsystems within the system. Subsystem n consists of mi components, and the subsystem state is the average of its components' states.

Structural metrics use inherent system characteristics as predictors for resilience, and their usefulness hinges on understanding of which characteristics serve as strong

predictors for system resilience. A risk in relying solely on structural metrics is that the proxy characteristics only weakly correlate to resilience and are more strongly correlated to other behaviors that are similar but distinct from resilience (such as robustness or reliability). While the use of structural metrics allows for resilience assessment without predefining disruption scenarios, a detailed model of the system is required to derive accurate values for the relevant properties.

Ade et al. [45] developed a methodology to quantify inherent safety at various stages of the design process given limited knowledge of the system. Given the process flow diagram of a potential process design, the inherent availability of the system can be calculated to represent the amount of time for which the system will be operational. Inherent availability is then included within the financial assessment to weigh profitability of each design against its availability.

The inherent availability of a system is given by

$$A_{inh,sys} = \prod_n^i A_{inh,i} \tag{8}$$

where

$$A_{inh,i} = \frac{MTTF_i}{MTTF_i + MTTR_i} \tag{9}$$

where n is the total number of equipment in the process system, $MTTF_i$ and $MTTR_i$ are the mean time to failure and mean time to repair of equipment i respectively, $A_{inh,i}$ is the inherent availability of equipment i, and $A_{inh,sys}$ is the inherent availability of the system (assuming all equipment are connected in series). The corrective maintenance downtime is evaluated through the following equation:

$$\frac{\text{corrective maintenance downtime}}{\text{total process time}} = 1 - A_{inh,sys} \tag{10}$$

The implementation of resilience enhancement strategies requires significant capital investment, which means the desire for more resilient systems needs to be balanced with economic considerations. Multi-objective formulations that aim to maximize resilience while minimizing costs have been proposed, as well as key performance indicators that modify economic metrics to explicitly include resilience indicators.

Al-Douri et al. [13] proposed a systematic methodology to consider failure during early conceptual design stages given a base design for the project. Bayesian updating procedure and Monte Carlo techniques are utilized to determine the appropriate distributions for the failure and repair scenarios, respectively. Next, Markov analysis is implemented to determine the system availability, which directly correlates to the process revenue function. Finally, the effects of failures are incorporated into the economic evaluation through the use of incremental return-of-investment (IROI) metric. The IROI describes the return of investment for choosing alternative design scenarios (additions or modifications of the base project), which can be maximized to determine the optimal design scenario given the failure and repair rates for each design and associated costs. The IROI is calculated using the following equation:

$$IROI = \frac{\Delta\text{Annual net profit of add}-\text{on project}}{\Delta\text{TCI of add}-\text{on project}} * 100 \tag{11}$$

To incorporate resilience into assessment of the financial viability of projects, Moreno-Sader et al. [46] proposed a novel Return-On-Investment (ROI)-based metric as a decision-making tool in the design of projects by extending the sustainability-weighted ROI metric [47] and sustainability and safety-weighted ROI metric [48] previously developed by adding resilience and reliability considerations. The metric can be used for a simulation-based analysis to evaluate the best-performing process alternative. The formulation for S2R2WROIM is given as follows:

$$\text{S2R2WROIM} = \frac{AEP_p\left[1 + \sum_{r=1}^{n}\sum_{s=1}^{m} w_{rs} w_{cr,s=1}\left(\frac{I_{base,rs} - I_{p,rs}}{I_{base,rs} - I_{t,rs}}\right)\right]}{TCI_p} \tag{12}$$

where AEP_p is the annualized economic profit; TCI_p is the funds needed to purchase and install the plant and is a function of fixed capital investment, working capital investment, and start-up investment; w_{rs} is the normalized weight for all the indicators; $w_{cr,s=1}$ is the weighing factor of categories; $I_{base,rs}$ is the indicator value for the base case; $I_{p,rs}$ is the indicator value of project p; $I_{t,rs}$ are the targeting values.

In quantifying process performance under uncertainty, system flexibility has been a pivotal indicator in the literature as it represents the feasible operational range of the system at steady state. Bruns et al. [49] provided a literature review of how the concept of flexibility has been applied in the chemical engineering field since the 1980s, when Grossmann and Sargent [50] published their work on optimization formulations for plant design with uncertain parameters. Furthermore, the authors defined four types of flexibility that target specific parts of a chemical system: feedstock, capacity, product, and operational flexibilities. Grossmann et al. [11] have outlined key mathematical formulations for analyzing the flexibility of an existing system as well as for designing new systems with a specified degree of flexibility. The flexibility analysis was incorporated into the synthesis of a process-intensified system by Tian and Pistikopoulos

[51] to guarantee a minimum level of flexibility in the optimal design. Rogers and Ierapetritou [18] applied the flexibility analysis to systems with black-box constraints, given stochastic uncertainties.

The dynamics of a process system span multiple spatial and timescales, as the system is affected by both the underlying physical phenomena involved and interactions with external agents such as the market, suppliers, and governments. This complexity inherent in process systems has led researchers to consider integrating the planning, scheduling, and control decisions involved in order to gain a more holistic understanding of process dynamics [52]. Resilience capabilities can therefore be built into the foundations of the system, from long-term design decisions to real-time control strategies. The scheduling of asset maintenance is also an important component of a comprehensive resilience strategy, as maintenance ensures the continued operations of existing equipment. Because maintenance makes up a significant portion of the operating budget of a process plant and also cuts into production runtime, efforts to cost-efficiently plan and schedule a process system need to account for the cost and time needed for maintenance [53].

Design and operations of resilient process systems

The utility of resilience assessments, whether qualitative or quantitative, is demonstrated when they are incorporated into process system design and operations. Including resilience considerations early in the design process is critical because more effective enhancement strategies can be implemented at a stage where system design is still amendable to change, especially strategies that focus on failure mitigation and adaptation. If resilience is not considered until after a detailed system design has been generated, it may be technically or financially challenging to make major changes. This concept is analogous to the concept of inherently safer design, where process safety is a vital consideration right from the beginning of system design [54]. Furthermore, the ability to compare multiple design alternatives on the basis of resilience capabilities is valuable to present optionality to decision-makers, especially when multiple objectives are pursued (e.g. economic, sustainability, etc.).

El-Halwagi et al. [55] outlined key insights for developing resilient designs of industrial plants through process integration. Strategies such as fail-safe design, utilization of modular, mobile, and repurposable components, and integrating redundancies and flexibility in system design have also been identified as enabling risk-averse and -resilient plant design.

Matelli and Goebel [56] devised a framework for evaluating design alternatives for cogeneration plants at the conceptual stage based on their resilience performance. Stochastic failure propagation simulations are run for each design option to assess its resilience to randomized disruptions to individual system components. The metrics used to evaluate resilience include the fraction of simulation runs that result in failure and the time to failure. Based on these metrics, the design that showed the best performance was chosen for further development. Running this assessment at the conceptual stage of process design gave decision-makers the information to eliminate underperforming alternatives before significant investments are made to develop these alternative designs further, and provided information on critical components that would need additional resources to protect them against failures.

da Silva and Matelli [57] built on this work by adding the possibility for repair actions on failed components. To implement this, a (random) probability of component repair is assigned to each component within the failure propagation simulation. In this representation, component repair is possible but not guaranteed. The addition of repair actions is shown to significantly increase the resilience performance of design alternatives, which could be a potential trade-off opportunity to explore further if repair actions are more cost-effective than building in redundancies into system design. Furthermore, the results of simulation runs provide guidance for future resource allocation toward repair of critical components.

Gong and You [21] compared the benefits of several resilience enhancement strategies on both resilience performance and capital cost of a shale gas processing plant through an adaptive robust optimization model. These strategies range from technology selection, adding redundancies to the system, and optimizing operational variables after disruption has occurred. The resilience metric chosen for this case study is the ratio of the quantity of products manufactured during disruptive events to the quantity without disruptive events. Based on the Pareto curve of optimal designs generated, the strategies most commonly deployed are building in redundancies (either through parallel processes or backup processes that only come online when the main process fails) or choosing the most resilient technology available.

To address the challenges of estimating the contribution of both qualitative and quantitative factors toward overall resilience performance, Zarei et al. [58] presented a fuzzy hybrid multi-objective decision-making model based on the relative importance of six main resilience indicators and 43 subindicators. The relative weight of each indicator was calculated using the fuzzy analytic hierarchy process (F-AHP), and then alternative designs were compared using the

fuzzy VIKOR method to select the most resilient design.

Ortiz-Espinoza et al. [59] developed a systematic integrated framework for explicitly including uncertainty in the performance evaluation stage of chemical process design. The uncertainty in the inputs is approximated using suitable probability functions and Monte Carlo simulations are carried out for sampling. Moreover, the conventional techno-economic analysis is extended to study the effect of uncertain input variables/parameters on process safety and environmental performance. This framework allowed for a range of performance outcomes to be generated along with their probabilities based on the likelihood of different parameter realizations, which can help inform the selection of design alternatives.

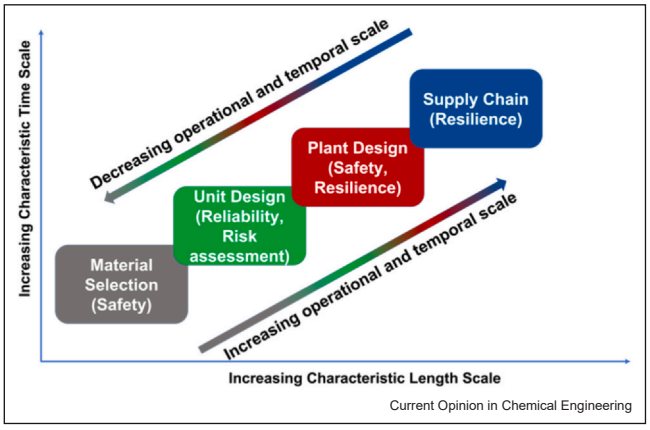
From this brief survey into the literature of resilience-aware design and operations, it can be seen that the integration of resilience assessment is frequently done at the conceptual design stage to screen potential design alternatives early or to decide between several resilience enhancement strategies to implement. The underlying assumption in these methodologies is that resilience is a response to disruptive events, so techniques such as stochastic simulations of failure or uncertain parameters are frequently used to represent disruptions to normal operations. The metrics used to evaluate design alternatives are overwhelmingly performance-based metrics, such as time to failure or performance level during disruptions. Trade-offs with key economic performance indicators, such as capital investment, need to be investigated as well to facilitate more comprehensive evaluations of design alternatives.

Research challenges

In pursuing an integrated approach to process system resilience, the following are a few open research opportunities. The scope of this work is limited to the technical aspects of process systems, namely process facilities and infrastructural elements of their supply chains (social and organizational factors are not considered at this time).

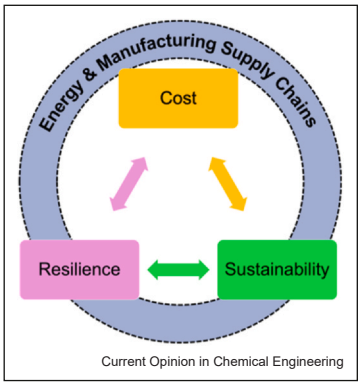
- *Multiscale integration:* A process system is not an isolated system: it makes up part of a larger supply chain and is itself composed of unit operations and materials (see Figure 2). All of these components operate on different spatial and temporal scales, and coordinating decisions across scales becomes important as decision-makers seek out more targeted, cost-competitive resilience enhancement strategies to implement [29]. Systematic techniques for molecular design, reaction pathway synthesis, process integration and synthesis, and supply chain management can be evolved to address resilience and to coordinate resilience measures at multiple scales. Analogous progress has been made in related areas such as safety, process intensification, and industrial symbiosis [54,60–62]. Furthermore, models with the right levels of detail will need to be developed at each scale and integrated among the multiple scales.
- *Holistic and multi-objective resilience framework:* The field of process design has seen similar but separate concepts develop such as process safety, flexibility, robustness, and reliability [63,64]. Uniting these concepts together under the banner of resilience can provide a more comprehensive and holistic view of how a process system behaves under disruptions of various levels of uncertainties. Conceptually, resilience is increasingly viewed as a cyclical process of preparation, adaptation, recovery, and continuous learning, so quantitative models will need to be developed to incorporate all four stages. Furthermore, multi-objective formulations will remain vital to balance economic and sustainability priorities with resilience [29] (Figure 3).
- *Resilience under uncertainty:* One major challenge in designing resilient process systems is the fact that future potential disruptions are not known for certain, and therefore systems cannot be designed for a specific disruption but for a range of possible scenarios. This uncertainty in the realization of design parameters can be represented in various ways, such as through the use of stochastic design parameters [65], simulation of extreme events [66], probabilistic network representations [39], or robust optimization [67]. Each method illuminates a different facet of resilience, and understanding of the business needs of the system will guide the choice of methodology for representing uncertainty. Future resilience frameworks need to include consideration of multiple sources of uncertainty to more faithfully represent real-life operating conditions.
- *Dynamic risk assessment and data-driven tools for resilience:* Dynamic risk assessment is the process of continuously updating the risk profile of a system based on real-time conditions to capture the ever-changing environment that the system operates in Ref. [68]. New data analytics, optimization, and simulation tools have significantly enhanced the fidelity of digital models of process systems, allowing decision-makers to have an accurate and dynamic view of the process state [69]. This increased visibility into the system opens up the possibility of an evolving resilience strategy based on the real-time conditions of the process [70].
- *Integration of technical, regulatory, and community aspects:* The challenges of designing and operating resilient process systems are not limited to technical aspects. Social resilience, which encompasses the community that lives and works around a process system, is an

Figure 2



Multiscale resilience analysis.

Figure 3



Integrated framework for next-generation data-driven cost-competitive energy and manufacturing supply chains.

equally important component of a holistic resilience framework [71]. During times of disruption, social resilience frameworks and policies become lifesaving measures that have the potential to reduce community suffering and even improve recovery performance of the system [72]. However, the difficulties in implementing social resilience strategies are distinct from the technical strategies we have focused on in this work [73].

Concluding remarks

In a volatile and uncertain world, it is crucial that the chemical industry protects itself and its assets from expected and unexpected disruptions. Building resilience capabilities within process systems in a cost-competitive manner is imperative to maintain stable and effective operations in the future. To accomplish cost-competitive resilience, important research thrusts to pursue include the integration of resilience assessments into process design and operations as early as is feasible, and the consideration of multiple spatial and temporal scales to uncover critical components and targeted enhancement strategies that are more cost-efficient.

To support resilience-aware design and operations, a comprehensive understanding of what constitutes resilience needs to be achieved. Multiple frameworks and metrics have been developed within the research community to target different aspects of resilience. However, work toward a unifying framework to quantify synergies and trade-offs among economics, sustainability, and resilience is still lacking. Exploration of how resilience interacts with other key business objectives needs to be pursued further to motivate investments in critical resilience enhancement strategies. Furthermore, process systems include social and organizational considerations on top of the technical aspects focused on in this work. Therefore, future resilience frameworks need to account for the role of social and organizational resilience that supports overall system performance.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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- • of special interest
- •• of outstanding interest.

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